Poster 1 — Who wants to be rich?

Topics 2.1-2.2 · Two-way tables, conditional relative frequencies, association

DOK 3

Your poster must show: the five "Rich" percents, a segmented bar chart (one 100% bar per superpower), your verdict, the repaired claim, and one association sentence.

Data (printed in the 2.4-2.6 follow-along)

Super Status! 433 high school students answered two questions: "What superpower would you choose?" and "What status would you prefer?"

Superpower	Famous	Happy	Healthy	Rich	Total
Fly	5	48	14	22	**89**
Freeze Time	3	63	9	26	**101**
Invisibility	3	62	11	22	**98**
Super Strength	4	43	17	13	**77**
Telepathy	0	49	0	19	**68**
Total	**15**	**265**	**51**	**102**	**433**

The claim you are judging

The student newsletter wrote: "Freeze Time people are the ones who want to be rich. 26 of them chose Rich, more than any other superpower."

DOK-3 — solve together before you build

a. For each superpower, find the percent of its fans who chose Rich. Write each as a fraction and a percent. Which total is the denominator each time, and why?

b. Draw a segmented bar chart: one bar per superpower, each bar 100% tall, split into Famous / Happy / Healthy / Rich. Add a key and a percent axis.

c. Verdict. Is the newsletter right about Freeze Time? Which superpower's fans are most likely to want to be rich, and by how many percentage points?

d. Repair. Rewrite the claim so it is true. Use a percent, not a count.

e. Is there an association between superpower and status? Compare two bars that look different (try Telepathy and Fly) and explain in one sentence.

Poster 1 — checklist, roles, sketch

Poster must-haves (= the rubric)

- ☐ Five "Rich" percents, each with its fraction and the row total as the denominator
- ☐ Segmented bar chart: five bars, each 100%, four segments, a key, a percent axis
- ☐ A verdict sentence that names the count AND the percent for Freeze Time and for the leader
- ☐ The repaired claim, true, with a percent
- ☐ One association sentence comparing two superpowers' bars

All five = E · three or four = P · fewer = I

The mistake this busts

In Unit 1 this class kept comparing counts from groups of different sizes: "the bigger count wins." Freeze Time has the most fans of any power (101), so of course it has a big Rich count. Percents *inside each row* are the fair comparison. On quiz 2.1, three of you also read a relative-frequency bar chart as if it showed counts.

Table Keeper
The five Rich fractions and percents, then all 20 percents for the bars. Show the arithmetic.

Visual
The segmented bar chart: five 100% bars, four segments, key, percent axis, title.

Writer
The verdict, the repaired claim, the association sentence. Use *given, conditional, association*.

Frame: "The claim compares _____ (counts). The probability it depends on is P(Rich | _____). Freeze Time: _____%. The leader is _____ at _____%. So the claim is _____ because _____."

given · conditional · relative frequency · association · denominator

Sketch the segmented bar chart here first (five bars, percent axis, key).

Draft the poster sentences here. Every sentence names the context and uses a number from the table.

Poster 2 — The flipped headline

Topic 2.6 · Conditional probability: order matters

DOK 3

Your poster must show: both conditional probabilities with their denominators, a tree diagram or two shaded tables, the verdict on each half of the headline, the repaired headline, and a second flipped pair.

Data (printed in the 2.4-2.6 follow-along)

Super Status! 433 high school students answered: "What superpower would you choose?" and "What status would you prefer?"

Superpower	Famous	Happy	Healthy	Rich	Total
Fly	5	48	14	22	**89**
Freeze Time	3	63	9	26	**101**
Invisibility	3	62	11	22	**98**
Super Strength	4	43	17	13	**77**
Telepathy	0	49	0	19	**68**
Total	**15**	**265**	**51**	**102**	**433**

The claim you are judging

A headline in the school paper: "Most students who want to be Famous chose Fly. So most students who chose Fly want to be Famous."

DOK-3 — solve together before you build

a. Find P(Fly | Famous) and P(Famous | Fly). Write each as a fraction with the correct denominator, then as a percent. Which total is the denominator each time?

b. Draw a tree diagram (first split Famous / not Famous, then Fly / not Fly) OR shade the "given" group on two copies of the table, to show why the denominators are 15 and 89.

c. Verdict on each half. Is "most" right for the first sentence? (How big is 5 out of 15?) Does the second sentence follow from the first?

d. Repair. Rewrite the headline as two true sentences, each with its probability.

e. Build. Find another pair in the table where flipping the "given" changes the answer a lot. Write both probabilities and explain in one sentence why they differ.

Poster 2 — checklist, roles, sketch

The flipped headline

Poster must-haves (= the rubric)

- ☐ Both conditionals as fractions with the right denominators (15 and 89) and as percents
- ☐ A tree diagram or two shaded tables that show the two "given" groups
- ☐ A verdict on each half: "most" judged against 50%, and the flip named
- ☐ The repaired headline: two true sentences with their numbers
- ☐ A second flipped pair from the table with both probabilities and a one-sentence why

All five = E · three or four = P · fewer = I

The mistake this busts

On 10/30 someone asked why P(B | A) and P(A | B) come out different. The word after "given" names the group you are standing inside; that group's total is the denominator. Flip the given, flip the denominator, and the answer changes. Your teacher's chocolate example: P(17 | likes chocolate) = 20/23 but P(likes chocolate | 17) = 20/25.

Table Keeper
The two fractions and percents, and the second flipped pair. Name each denominator.

Visual
The tree (two levels, probabilities on every branch) or the two shaded tables.

Writer
The verdict on each half, the repaired headline, the why sentence. Use *given, conditional*.

Frame: "P(Fly | Famous) means: of the _____ students who want to be Famous, _____ chose Fly = _____%. P(Famous | Fly) means: of the _____ students who chose Fly, _____ want to be Famous = _____%. The headline flips _____, so _____."

given · conditional · denominator · plurality · majority

Sketch the tree or the two shaded tables here first.

Draft the poster sentences here.

Poster 3 — Can't happen together, so unrelated?

Topics 2.5 + 2.7 · Mutually exclusive vs independent

DOK 3

Your poster must show: the joint probability, two independence tests, a verdict that uses both words, the general argument, and two 2x2 tables you designed.

Data (printed in the 2.4-2.6 follow-along)

Super Status! 433 high school students answered: "What superpower would you choose?" and "What status would you prefer?"

Superpower	Famous	Happy	Healthy	Rich	Total
Fly	5	48	14	22	**89**
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Total	**15**	**265**	**51**	**102**	**433**

The claim you are judging

"Nobody who chose Telepathy wants to be Famous: 0 of 68. Since the two never happen together, Telepathy and Famous are independent. One has nothing to do with the other."

DOK-3 — solve together before you build

a. Are "chose Telepathy" and "wants to be Famous" mutually exclusive? Show the joint probability P(Telepathy and Famous).

b. Are they independent? Test two ways: compare P(Famous) with P(Famous | Telepathy); compare P(Famous) × P(Telepathy) with P(Famous and Telepathy).

c. Verdict, in one sentence that uses both terms correctly. What does knowing "Telepathy" do to the chance of "Famous"?

d. Prove it in general. Two events can each happen (each has probability above 0) and they are mutually exclusive. Explain why they can never be independent. (Hint: what does independence say P(A and B) must equal?)

e. Build. Make a 2x2 table of 100 students for "plays a sport" and "plays an instrument" where the two events are independent but NOT mutually exclusive (show the check). Then make a second table where they are mutually exclusive, run the independence check again, and say what you find.

Poster 3 — checklist, roles, sketch

Can't happen together, so unrelated?

Poster must-haves (= the rubric)

- ☐ P(Telepathy and Famous) = 0 shown from the table, with the words *mutually exclusive*
- ☐ Independence tested two ways with the numbers (P(Famous) vs P(Famous | Telepathy); product vs joint)
- ☐ One verdict sentence that uses *mutually exclusive* AND *independent* correctly
- ☐ The general argument from part (d), written as a short proof that starts from what independence says P(A and B) must equal
- ☐ Two 2x2 tables (100 students each) with the independence check written under each

All five = E · three or four = P · fewer = I

The mistake this busts

On 10/29 the class asked for the difference between mutually exclusive and independent, and your teacher promised the class an example that settles it. Is "can't happen together" the same as "unrelated"? Your table decides, with numbers: test what knowing one event does to the probability of the other.

Table Keeper
The four probabilities, the product, and the numbers in both 2x2 tables.

Visual
The two 2x2 tables (must-have 5), each with its check written underneath; add a Venn diagram of the two events if there is room.

Writer
The verdict sentence and the general argument. Use *mutually exclusive, independent, joint*.

Frame: "P(Telepathy and Famous) = _____, so they are _____. But P(Famous) = _____ and P(Famous | Telepathy) = _____, which are (equal / not equal), so they (are / are not) _____. Knowing a student chose Telepathy _____ the chance they want to be Famous."

mutually exclusive | independent | joint | given | product

Sketch the Venn diagram and the two 2x2 tables here first.

Draft the poster sentences here.

Poster 4 — Double counted

Topics 2.4 + 2.7 · Unions, complements, independence

DOK 3

Your poster must show: a Venn diagram (or two-way table) with all four regions, the corrected union, two independence tests, the sentence on why the two tests agree, and the "exact independence" number.

Data (printed in the 2.7 follow-along, from the 2010 AP exam)

A survey of 2,500 adults asked about education level and where they get their news.

Group	Count
College graduates	693
Get news mainly from the internet	687
College graduates who get news from the internet	245
All adults surveyed	**2,500**

The claim you are judging

A blog wrote: "P(college grad or internet news) = 693/2500 + 687/2500 = 55.2%. And college grads are no more likely than anyone else to get their news online."

DOK-3 — solve together before you build

a. Draw a Venn diagram (or a two-way table) and fill in all four regions: both, grad only, internet only, neither. Check that they add to 2,500.

b. Find the mistake in the 55.2%. Compute the correct P(grad or internet). Which people were counted twice?

c. Test the second claim two ways: compare P(internet | grad) with P(internet); compare P(grad) × P(internet) with P(grad and internet). Verdict: are the events independent? Who is more likely to read news online?

d. Explain in one sentence why the two tests in (c) must always agree. (Hint: start from P(A and B) = P(A) × P(B) and divide both sides by P(A).)

e. Build. Keep 693 college grads and 687 internet readers. What number in "both" would make the two events exactly independent? Is it possible with real people? What does that tell you?

Poster 4 — checklist, roles, sketch

Double counted

Poster must-haves (= the rubric)

- ☐ Four regions filled and summing to 2,500 (both / grad only / internet only / neither)
- ☐ The group that was counted twice named, and the corrected union as a fraction of 2,500 and a percent
- ☐ Both independence tests with numbers, and a verdict naming who is more likely
- ☐ One sentence on why the two tests always agree
- ☐ The "both" count that would make the two events exactly independent, and what it tells you

All five = E · three or four = P · fewer = I

The mistake this busts

2.7 (Independent Events and Unions) is the lesson this class finished least. Its exit ticket set two traps, and both are hiding in this claim: one in the first sentence's arithmetic, one in how the second sentence was checked. Find both.

Table Keeper
The four regions, the union, both tests, the exact-independence count. Show every subtraction.

Visual
The Venn diagram with the overlap shaded (or the 2x2 table), every region labeled with its count.

Writer
The double-count sentence, the verdict, the why-the-tests-agree sentence. Use *union, joint, independent*.

Frame: "The 55.2% went wrong because _____. The correct P(grad or internet) = _____. P(internet | grad) = _____ and P(internet) = _____, so the events are _____, and _____ are more likely to read news online."

union · joint · independent · given · complement

Sketch the Venn diagram here first (two circles, four regions, counts).

Draft the poster sentences here.

Poster 5 — Is the game fair?

Topics 2.8, 2.9 + 2.3 · Random variables, mean and SD, simulation

DOK 3

Your poster must show: the probability distribution of net profit, the mean and standard deviation with work, both interpretations, your verdict, the fair fix, and your simulation average.

Data (the 2.8 follow-along exit ticket)

A carnival game costs $2 to play. You roll a fair six-sided die. Roll a 6: you win $10. Roll a 1: you win $3. Otherwise: you win nothing.

The claim you are judging

The sign on the booth: "A FAIR GAME! You win on 2 of every 6 rolls."

DOK-3 — solve together before you build

a. Let X = the player's net profit for one game (winnings minus the $2). List every value of X with its probability in a table. Check that the probabilities add to 1.

b. Compute the mean μ_X and the standard deviation σ_X. Show the formula, the substitution, and the arithmetic.

c. Interpret μ_X and σ_X in context. Use "in the long run" and say "per game".

d. Verdict. Is the game fair? For whom? The sign's "2 of every 6 rolls" is really a 2/6 chance on each roll, not a promise: why is it still the wrong number to look at? Who does the sign fool, and who does the game actually favor?

e. Repair. Change ONE payout so that $\mu_X = 0$ exactly. Show the new table and the new mean.

f. Simulate. On the TI-84, randInt(1,6) is one game. Play 20 games, record the net each time, and find your average. Write it next to μ_X. Why do they differ, and what would happen with 1,000 games?

Poster 5 — checklist, roles, sketch

Poster must-haves (= the rubric)

- ☐ Distribution table of X = net profit (not winnings) with probabilities that sum to 1, plus a probability histogram of X
- ☐ μ_X and σ_X with the formula, the substitution and the arithmetic shown
- ☐ Both interpretations in context, with "in the long run" and "per game"
- ☐ A verdict naming who the game favors and why "2 of 6" is the wrong number
- ☐ The one-payout fix with the new mean shown, and the dotplot of your 20 simulated games with its average marked next to μ_X

All five = E · three or four = P · fewer = I

The mistake this busts

Three traps from the 2.8 exit ticket: using the winnings instead of the NET (winnings minus the $2 you paid); reading the mean as what happens in every single game; and calling a game "fair" without asking fair to whom. And from 10/26: a simulation estimates; it does not give the exact answer.

Table Keeper
The table, μ, σ, the fix. Show the formula before the numbers.

Visual
A probability histogram of X, and a dotplot of your 20 simulated nets with their average marked.

Writer
The interpretations and the verdict. Use *expected value, long run*.

Frame: "X = _____ with probabilities _____. μ_X = _____, which means in the long run the player _____ about $_____ per game. σ_X = _____, so a single game typically ends about $_____ from the mean. The game favors the _____ because _____."

expected value · long run · net · fair · simulation

Sketch the probability histogram and the simulation dotplot here first. TI-84: MATH → PROB → randInt(1,6).

Draft the poster sentences here.

Poster 6 (spare) — Rigged coin?

Topic 2.3 · Estimating probabilities using simulation

DOK 3

Your poster must show: the simulation design, 25 recorded trials, a dotplot of the longest runs, the estimated probability, and your verdict.

Data

You make the data: 20 flips of a fair coin, simulated on the TI-84 with randInt(0,1,20), where 1 = heads. (MATH → PROB → randInt. ENTER repeats it.)

The claim you are judging

"*I flipped a coin 20 times and got 6 heads in a row. That can't happen with a fair coin. It's rigged.*"

DOK-3 — solve together before you build

a. Design. Write the simulation in three sentences: how one flip is modeled, what one trial is, and what you record from each trial (the longest run of heads in the 20 flips).

b. Run 25 trials. Record the longest run each time. Draw a dotplot of the 25 values.

c. Estimate P(longest run is 6 or more) from your dotplot.

d. Verdict. Is 6 in a row evidence of a rigged coin? Use your estimate.

e. Your estimate is not the exact answer. Why not? What would 1,000 trials do? (Your teacher has the exact value.)

Poster 6 — checklist, roles, sketch

Poster must-haves (= the rubric)

- ☐ The design in three sentences (flip, trial, what is recorded)
- ☐ 25 trials recorded in a table
- ☐ A dotplot of the 25 longest runs with a scale and a title
- ☐ The estimated probability as a fraction of 25 and a decimal
- ☐ The verdict, plus one sentence on what more trials would do

All five = E · three or four = P · fewer = I

The mistake this busts

On 10/26 someone asked whether the simulated coin was fair. A simulation estimates a probability from many trials; it never proves a single outcome impossible. Rare is not impossible, and one surprising result is not proof of rigging.

Table Keeper
Runs the calculator, records the 25 longest runs, computes the fraction.

Visual
The dotplot with a scale that covers every value you got (expect 2 to 8; 0 and 20 are possible) and the "6 or more" region shaded.

Writer
The design sentences and the verdict. Use *simulation, relative frequency, long run*.

Frame: "One trial is _____. We recorded _____. In _____ of 25 trials the longest run was 6 or more, so we estimate P(run of 6 or more) ≈ _____. A run of 6 is _____, not _____, so the coin _____."

simulation · trial · relative frequency · long run · estimate

Record the 25 trials and sketch the dotplot here first.

Draft the poster sentences here.